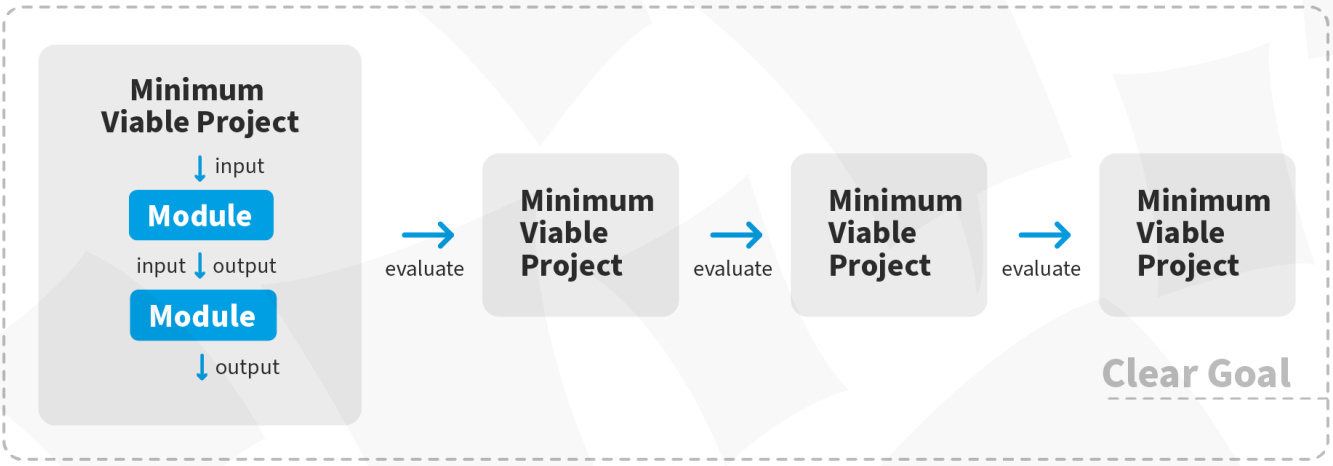

How to Apply Incremental Modular Design to Global Challenges

Incremental modular design flexibly breaks large-scale projects into small stages or modules. Every module is developed and tested on its own before integration. This allows for easier maintenance and a more scalable solution. Each module builds on the other to solve large-scale problems, piece by piece. Incremental modular design combines incrementalism with concepts borrowed from project management and computer science. With this framework, you can pitch and secure funds for large projects and retain the flexibility of several smaller incremental steps.

Step 1: Define the Goal

All stakeholders must agree to the goals of the larger project. Goals align the team and provide direction. At the same time, the goals help bring sponsors on board even if the exact details of the project are not clear.

Clear Goal

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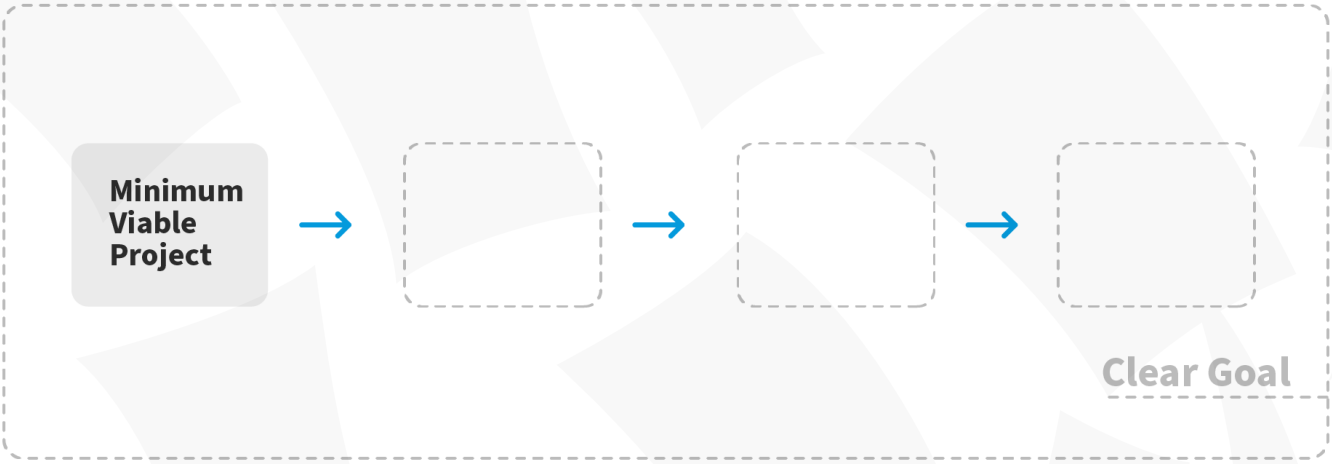
Step 2: Divide the Solution into Smaller Steps

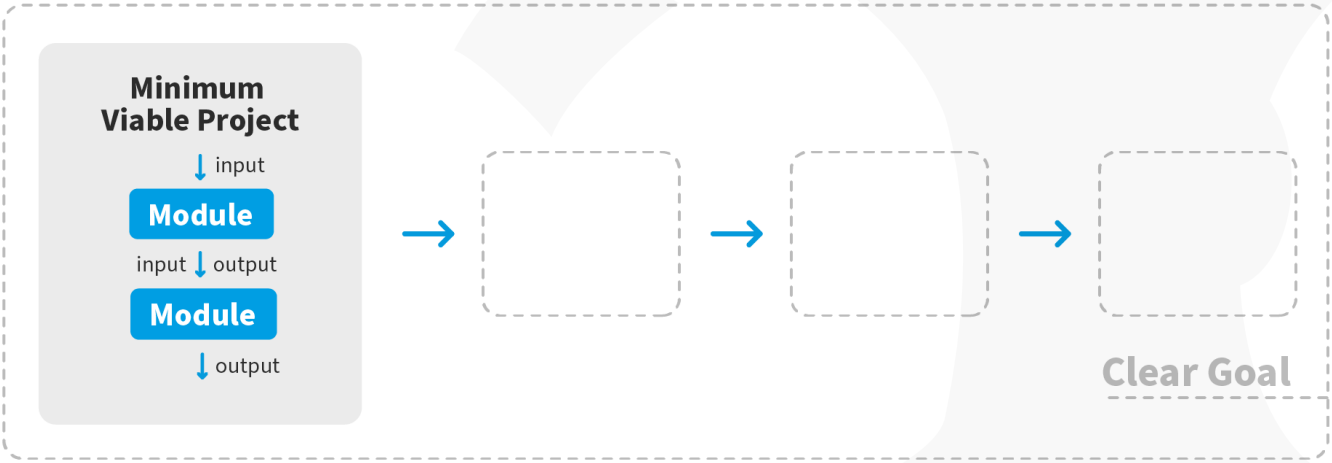
Work towards the larger goal through smaller steps. The idea is to not develop a massive project plan with every minute detail defined upfront. At this stage, you will have a bird's-eye view of the entire solution.



Step 3: Define Small Independent Modules within a Project

Break your projects down into even smaller modules. The key requirements for the modules are:

- Each module must be the **smallest** size that can produce a **useful result**.
- Clearly state the **input and output** requirements for each module. The output of one module becomes the input for the next module.
- It doesn't matter how each module works internally, as long as they meet the requirements of the inputs and outputs.



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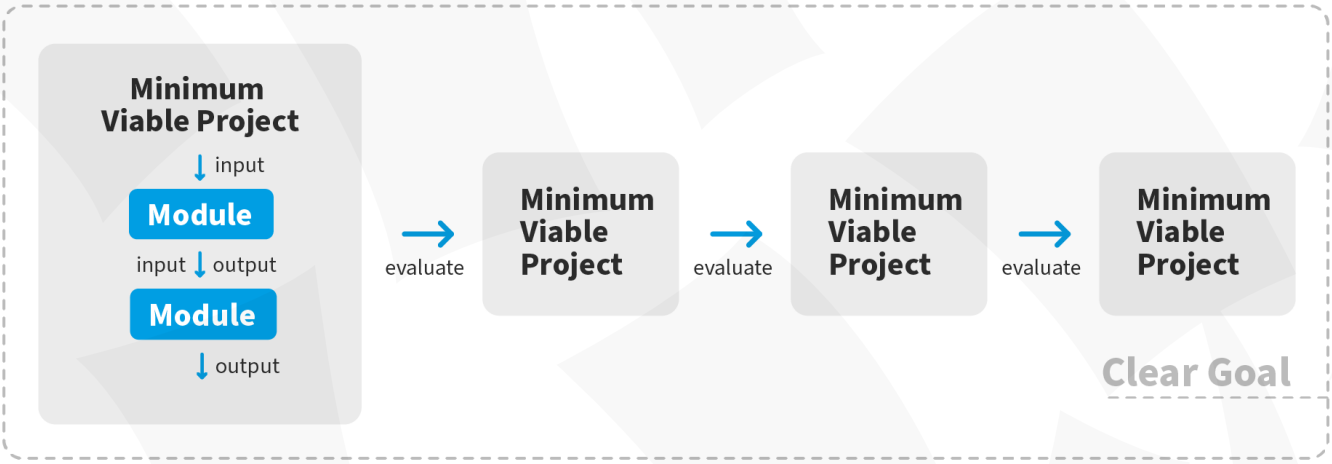
Step 4: Deliver, Evaluate and Move Forward

Deliver each module, evaluate it and decide the next course of action based on the analysis. Since the modules are small and useful, you can deliver results fast and tweak your solution if the module (and the project) do not deliver the expected results.



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